

NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MASTER'S in COMPUTER SCIENCE & ENGINEERING

MASTER'S DEGREE THESIS
IN
CLOUD COMPUTING

Predictive Energy-Aware Auto-Scaling for Cost-Efficient Cloud Resource Management

MENTOR:
Prof. Manoj Kumar Patra

CANDIDATES:
Hrishikesh Chandra - 26CSM1R05
Prakash Kumar - 26CSM1R16

ACADEMIC YEAR 2026–2027

Table of Contents

Abstract 1

Problem Statement 2

Proposed Solution 3

Overview 3

System Architecture 4

Detailed Working of the Proposed System 5

Flowchart of the Proposed Algorithm 8

Datasets 10

Dataset Description 10

Evaluation Metrics 11

Conclusions 13

Project Summary 13

References 14

Abstract

Predictive Energy-Aware Auto-Scaling for Cost-Efficient Cloud Resource Management

Cloud computing provides scalable and flexible computing resources to meet the varying demands of applications. However, inefficient resource allocation can lead to increased operational costs and energy consumption, particularly when resources are over-provisioned or under-utilized. Traditional reactive auto-scaling approaches adjust resources based on current system metrics, which may not respond efficiently to sudden changes in workload. To address these challenges, this project proposes a predictive energy-aware auto-scaling approach for cost-efficient cloud resource management.

The proposed system predicts future workload from historical utilization data and scales resources proactively. By factoring in energy consumption and resource utilization, it aims to maintain performance while reducing resource wastage, energy usage, and cost. It will be evaluated on prediction accuracy, response time, energy consumption, and operational cost, and compared against traditional reactive auto-scaling.

The expected result is a more efficient cloud resource management framework that balances performance, energy efficiency, and cost. This approach can contribute to sustainable cloud computing by reducing resource wastage while maintaining the required quality of service.

Keywords: Cloud Computing, Predictive Auto-Scaling, Energy Efficiency, Resource Management, Workload Prediction, Cost Optimization.

Problem Statement

Existing auto-scaling mechanisms in cloud computing environments primarily rely on reactive approaches, where resources are scaled up or down only after predefined resource utilization thresholds are crossed. This approach presents several challenges:

- **Delayed response to workload fluctuations:** Sudden increases in demand may cause increased response time, resource contention, and potential Service Level Agreement (SLA) violations.
- **Resource over-provisioning:** Allocating more resources than required during low-demand periods leads to under-utilization and unnecessary cloud expenditure.
- **Excessive energy consumption:** Underutilized computing resources may continue consuming significant energy, resulting in inefficient power utilization.
- **Lack of workload prediction:** Traditional reactive approaches do not adequately anticipate future resource requirements, limiting proactive resource allocation.
- **Trade-off between cost, energy, and performance:** Optimizing cloud expenditure and energy consumption while maintaining application performance remains a challenging task.

Consequently, the absence of an integrated predictive and energy-aware auto-scaling mechanism can lead to increased operational costs, inefficient resource utilization, unnecessary energy consumption, and degraded application performance during workload fluctuations. Therefore, an intelligent auto-scaling approach is required to predict future workload demands and dynamically allocate cloud resources while balancing cost efficiency, energy consumption, and Quality of Service (QoS).

Proposed Solution

Overview

The proposed system aims to develop an intelligent auto-scaling mechanism that predicts future workload demands and adjusts cloud resources proactively. Unlike traditional reactive auto-scaling, which responds after resource utilization crosses a predefined threshold, the proposed approach uses historical workload data to anticipate upcoming demand and make scaling decisions in advance.

The system integrates three major objectives:

1. **Workload prediction:** Forecast future CPU utilization or incoming workload using historical monitoring data.
2. **Energy-aware resource allocation:** Allocate only the resources required to handle the predicted workload while reducing unnecessary energy consumption.
3. **Cost-efficient auto-scaling:** Dynamically add or remove virtual machines (VMs) or containers while maintaining acceptable response time and Quality of Service (QoS).

The proposed solution will be evaluated against a conventional reactive auto-scaling approach to determine whether predictive scaling can improve resource utilization, reduce energy consumption, and lower operational costs.

System Architecture

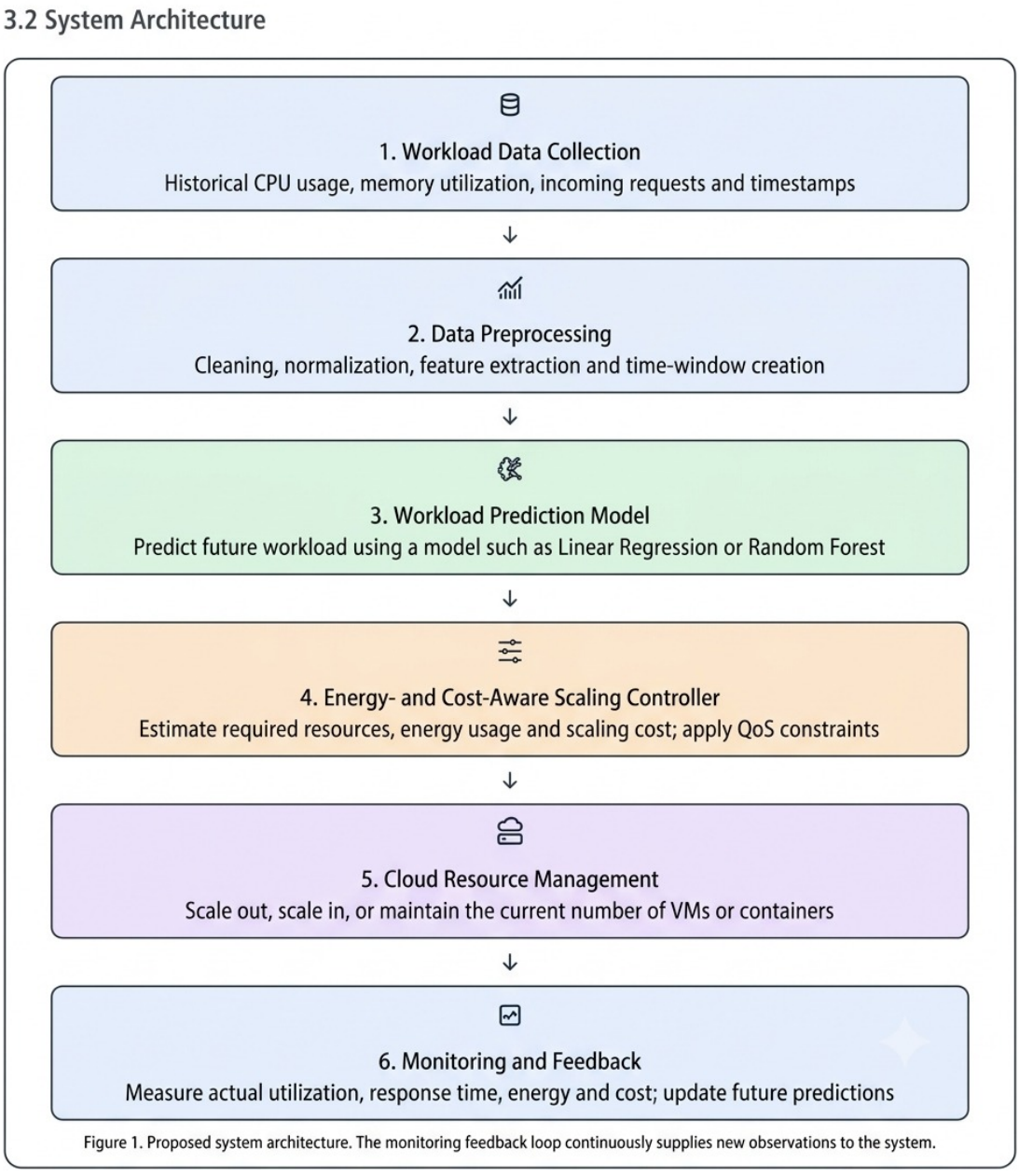


Figure 1. Proposed system architecture. The monitoring feedback loop continuously supplies new observations to the system.

The architecture follows a closed-loop approach. The prediction model estimates upcoming demand, the scaling controller decides how many resources are needed, and the monitoring component checks whether the decision achieved the desired performance.

Detailed Working of the Proposed System

1. **Step 1: Workload Data Collection**

   The system collects historical and real-time data from cloud workloads. The dataset may contain:

   - CPU utilization (%)
   - Memory utilization (%)
   - Number of incoming requests
   - Response time (ms)
   - Number of active VMs or containers
   - Estimated power consumption (W)

2. **Step 2: Data Preprocessing**

   The collected data is cleaned and arranged in chronological order. Missing values, duplicate records, and invalid measurements are handled before training the prediction model.

   For example, the model may use CPU utilization from the previous five time intervals to predict utilization in the next interval.

   Table 1: Historical CPU Utilization and Target Values

Time Interval	**Previous CPU Values (%)**	**Target CPU (%)**
T1	30, 32, 35, 38, 40	43
T2	32, 35, 38, 40, 43	46
T3	35, 38, 40, 43, 46	50

   Note: The values shown are illustrative examples and are not measured project results.

3. **Step 3: Predict Future Workload**

   A machine learning model predicts future CPU utilization or request demand based on historical data.
   For a manageable implementation, Linear Regression can be used as the baseline predictor. It can be compared with Random Forest if time permits.

For example, if current CPU utilization is 55% but the model predicts 85% in the next interval, the controller can scale out before the workload spike occurs.

4. **Step 4: Calculate the Required Resources**

   The system estimates the number of virtual machines (VMs) required to handle the predicted workload while maintaining the target CPU utilization.

   The required number of VMs can be estimated using:

$$N_{\text{required}} = \left\lceil \frac{U_{\text{predicted}}}{U_{\text{target}}} \right\rceil$$

   where both utilization values must represent workload demand on the same single-VM capacity scale.

   For example, if the predicted workload is equivalent to 170% of the capacity of one VM and the target utilization is 85%, then:

$$N_{\text{required}} = \left\lceil \frac{170}{85} \right\rceil = 2$$

   The scaling controller compares the required number of VMs with the currently active VMs and takes one of the following actions:

   - **Scale Out:** Add VMs when more resources are required.
   - **Scale In:** Release unnecessary VMs when demand is expected to remain low and performance requirements are satisfied.
   - **Maintain:** Keep the current allocation when it matches the estimated requirement.

   The controller also considers minimum and maximum VM limits, VM startup delays, and application performance requirements to avoid unnecessary scaling actions and maintain the required Quality of Service (QoS).

5. **Step 5: Apply Energy- and Cost-Aware Decisions**

   The controller evaluates possible scaling decisions against three objectives:

- **Cost:** Minimize the cost of running VMs or containers.
- **Energy:** Reduce unnecessary power consumption by avoiding idle resources where possible.
- **Performance:** Keep response time and SLA violations within acceptable limits.

A simplified optimization objective is:

$$J = w_c C + w_e E + w_p P$$

where:

- C = normalized cloud cost
- E = normalized energy consumption
- P = normalized performance penalty, such as SLA violations
- w_c, w_e, w_p = weights assigned to the three objectives

The controller selects a scaling action that minimizes J, subject to minimum resource requirements and performance constraints. Normalization is important because cost, energy, and latency use different units.

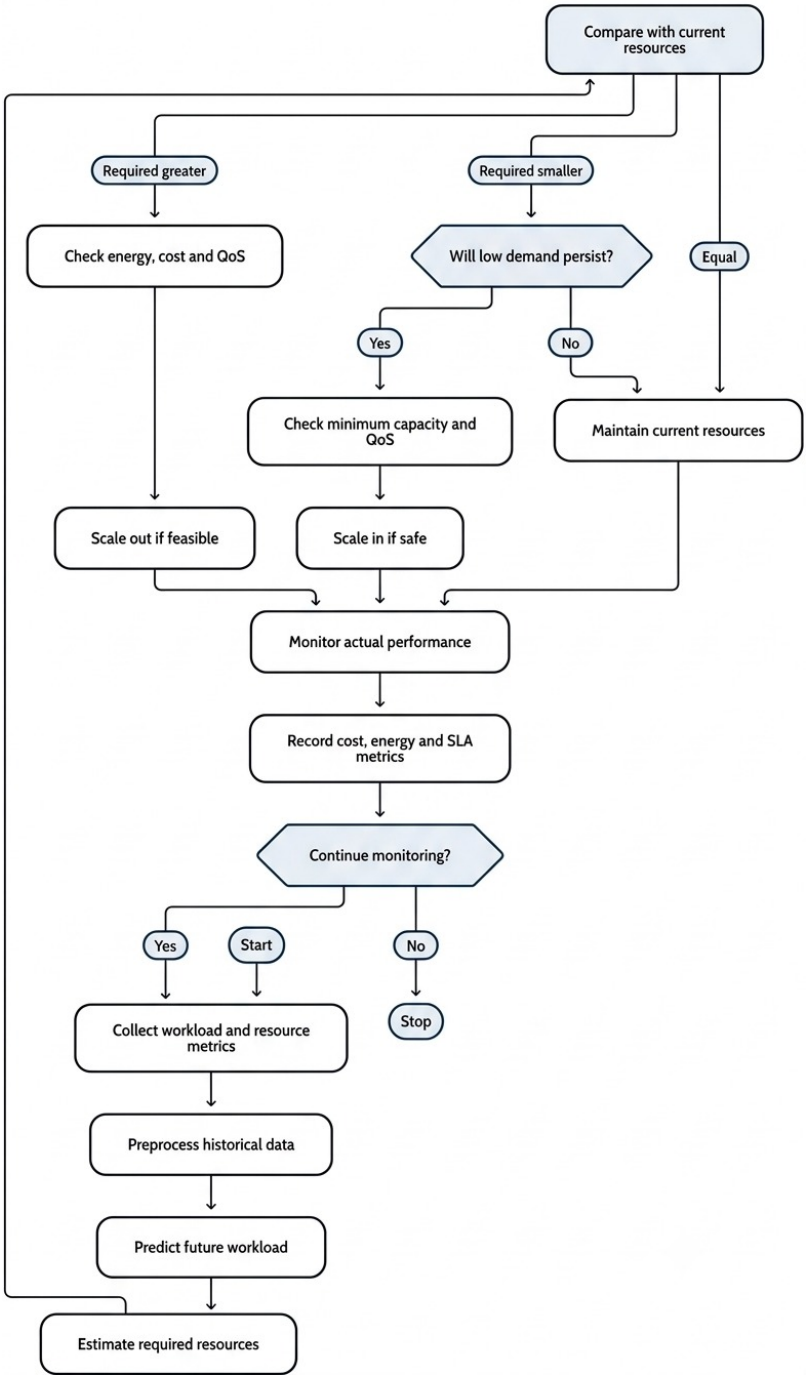
6. **Step 6: Monitor and Evaluate**

   After applying the scaling decision, the system records actual workload, CPU utilization, response time, energy consumption, and cost. These observations are fed back into the next prediction and scaling cycle.

   This feedback loop helps identify prediction errors and determine whether the proposed scaling policy performs better than the reactive baseline.

Flowchart of the Proposed Algorithm

The following flowchart illustrates the complete decision-making process, including the scale-out and scale-in decisions.



Flowchart interpretation: Prediction provides advance warning of workload changes. The controller then checks whether scaling is needed and ensures that cost or energy savings do not compromise application performance. Scale-in is handled more cautiously than scale-out because removing resources too early may cause overload

Datasets

Dataset Description

This project utilizes workload traces obtained from the official Grid Workloads Archive (GWA), a publicly available repository of anonymized workload data collected from grid computing environments. The archive provides real-world workload traces that support research on workload characterization, resource management, and scheduling.
The selected trace will be analyzed to understand workload arrival patterns and resource requirements. The data will be preprocessed to handle missing or invalid values and to extract relevant features for workload prediction. Based on the available attributes, the workload prediction model will forecast future demand, which will subsequently be used by the proposed energy-aware auto-scaling controller.
The performance of the proposed approach will be evaluated against a reactive scaling baseline using appropriate metrics, including resource utilization, operational cost, estimated energy consumption, and workload response or completion time where supported by the experimental setup.

Dataset Source: Grid Workloads Archive (GWA),

Link to the Dataset

Evaluation Metrics

The proposed system will be evaluated using the following metrics to compare predictive auto-scaling with traditional reactive auto-scaling.

1. **Mean Absolute Error (MAE):** Measures the average difference between actual and predicted workload. Lower MAE indicates better prediction accuracy.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{1}$$

2. **Resource Utilization:** Measures the percentage of allocated computing capacity being used.

$$\text{Resource Utilization (\%)} = \frac{\text{Used Capacity}}{\text{Total Allocated Capacity}} \times 100 \tag{2}$$

3. **Energy Consumption:** Measures the total energy consumed by computing resources.

$$\text{Total Energy (kWh)} = \frac{\sum[\text{Power (W)} \times \text{Time (hours)}]}{1000} \tag{3}$$

4. **Operational Cost:** Calculates the total cost of allocated cloud resources during the experiment.

$$\text{Total Cost} = \sum(\text{Resource Cost per Hour} \times \text{Usage Duration in Hours}) \quad (4)$$

5. **Average Response Time:** Measures the average time taken to process requests.

$$\text{Average Response Time} = \frac{\sum \text{Response Time of All Requests}}{\text{Total Number of Requests}} \quad (5)$$

6. **SLA Violation Rate:** Measures the percentage of requests that fail to meet the specified Service Level Agreement (SLA).

$$\text{SLA Violation Rate (\%)} = \frac{\text{Number of SLA Violations}}{\text{Total Number of Requests}} \times 100 \quad (6)$$

7. **Cost and Energy Reduction:** Measures the percentage reduction in cost or energy consumption compared with reactive auto-scaling.

$$\text{Reduction (\%)} = \frac{\text{Baseline Value} - \text{Proposed Value}}{\text{Baseline Value}} \times 100 \quad (7)$$

Note: Energy, cost, and response-time metrics require suitable measurements or documented estimation models in addition to the GWA workload trace.

Conclusions

Project Summary

This project proposes a predictive energy-aware auto-scaling approach for cost-efficient cloud resource management. The proposed system uses workload traces obtained from the official Grid Workloads Archive (GWA) to analyze workload patterns and predict future resource demands. The collected data is preprocessed and used to train a workload prediction model, which estimates upcoming demand and supports proactive resource allocation.

Based on the predicted workload, the scaling controller determines whether to scale out, scale in, or maintain the current number of virtual machines. The controller considers resource utilization, energy consumption, operational cost, and application performance to make efficient scaling decisions. A monitoring and feedback mechanism continuously evaluates actual system behavior and provides updated information for subsequent scaling decisions.

The system will be evaluated by comparing predictive auto-scaling with traditional reactive auto-scaling using metrics such as Mean Absolute Error (MAE), resource utilization, total energy consumption, operational cost, average response time, SLA violation rate, and cost and energy reduction. Where direct measurements are unavailable in the GWA trace, energy consumption and operational cost will be estimated using documented assumptions.

Overall, the proposed approach aims to improve resource utilization, reduce unnecessary energy consumption and operational expenditure, and maintain acceptable application performance. The experimental evaluation will determine the effectiveness and practical limitations of predictive auto-scaling for efficient and sustainable cloud resource management.

References

- iosup2008 A. Iosup, H. Li, M. Jan, S. Anoep, C. Dumitrescu, L. Wolters, and D. H. J. Epema, "The Grid Workloads Archive," *Future Generation Computer Systems*, vol. 24, no. 7, pp. 672–686, 2008.
  DOI: https://doi.org/10.1016/j.future.2008.02.003.

- moreno2019 R. Moreno-Vozmediano, R. S. Montero, E. Huedo, and I. M. Llorente, "Efficient resource provisioning for elastic Cloud services based on machine learning techniques," *Journal of Cloud Computing*, vol. 8, article 5, 2019.
  DOI: https://doi.org/10.1186/s13677-019-0128-9.

- kumari2022 M. Kumari, G. Sahoo, K. K. Senapati, and G. Kumar, "Energy-aware autoscaling for scientific workflow in cloud environment," *Concurrency and Computation: Practice and Experience*, vol. 34, no. 3, article e6576, 2022.
  DOI: https://doi.org/10.1002/cpe.6576.

- xue2022 S. Xue et al., "A Meta Reinforcement Learning Approach for Predictive Autoscaling in the Cloud," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 4290–4299, 2022.
  DOI: https://doi.org/10.1145/3534678.3539063.

- aldossary2018 M. Aldossary and K. Djemame, "Performance and Energy-Based Cost Prediction of Virtual Machines Auto-Scaling in Clouds," in *Proceedings of the 44th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*, 2018.
  DOI: https://doi.org/10.1109/SEAA.2018.00086.